

STEVEN DILLMANN

PHD CANDIDATE · STANFORD UNIVERSITY · AI FOR SCIENCE

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EDUCATION

Stanford University

PhD Computational & Applied Mathematics (ICME)

Sep 2024 – Present
Stanford, CA, USA

Research: AI for Science, AI Evaluation, Agents for Science; currently leading [Terminal-Bench-Science](#)

Advisors & Faculty Collaborators: Sanmi Koyejo (CS), Risa Wechsler (Physics), Ludwig Schmidt (CS)

Fellowships: Stanford Interdisciplinary Graduate Fellowship (SIGF), Stanford HAI Data Science Scholar

University of Cambridge, St John's College

MPhil Data Intensive Science

Oct 2023 – Jul 2024
Cambridge, UK

Grade: Distinction (87.7%), ranked 1st of 62

Thesis: Deep Learning for the Prompt Detection of Gravitational Wave Signals (97%, highest mark in cohort)

Coursework: Machine Learning, Deep Learning, Bayesian Statistics, Software Engineering

Imperial College London

MEng Aeronautics with Spacecraft Engineering

Oct 2018 – Sep 2023
London, UK

Grade: First Class Honours (83.2%), ranked 1st of 132 (Head of Department Award), Dean's List 2019–2023

Thesis: Representation Learning for the Discovery of Astrophysical X-ray Transients (90%, Best Computational Thesis Award)

Visiting Research: California Institute of Technology, Harvard University

EXPERIENCE

Stanford Artificial Intelligence Laboratory (SAIL)

Graduate Researcher

Oct 2024 – Present
Stanford, CA, USA

- Leading [Terminal-Bench-Science](#), which extends [Terminal-Bench](#) (ICLR 2026) to evaluate AI agents on scientific research
- Developing a generative model that predicts galaxy emission across multi-wavelength, multi-scale observations
- Designed agent-evaluation tasks for [ReplicationBench](#) and built its adapter for [Harbor Adapters](#) (NeurIPS D&B 2026)
- Designed biosafety and bioethics evaluations for the genomic foundation model [Evo 2](#) (Nature 2026)
- Co-authored the “AI for Science” chapter of the Stanford HAI [2026 AI Index Report](#)

Allen Institute for AI (AI2), Asta Team

Research Intern

Jun 2026 – Sep 2026
Seattle, WA, USA

- Built AstaVerse, an agentic multiverse-analysis system that tests hypotheses and how analysis choices shift their outcomes
- Developed AstaVerse's optional human-in-the-loop interface to upload datasets, pose hypotheses, and review decisions

SLAC National Accelerator Laboratory

Research Assistant

Jan 2025 – Apr 2025
Stanford, CA, USA

- Trained normalizing flows for simulation-based inference of cosmological parameters from strong gravitational lensing
- Wrote a neural posterior estimation tutorial for [SLSim](#), the Rubin Observatory LSST strong gravitational lensing simulator

Harvard-Smithsonian Center for Astrophysics, Harvard University

Visiting Research Student

Jan 2023 – Jun 2023
Cambridge, MA, USA

- Developed a representation learning method for anomaly detection and similarity search in Chandra X-ray data (MNRAS 2025)
- Discovered [XRT 200515](#) with this method, the first extragalactic fast X-ray transient of its kind in the Chandra archive

NASA Jet Propulsion Laboratory, California Institute of Technology

Machine Learning Intern

Jun 2022 – Oct 2022
Pasadena, CA, USA

- Designed a density-based clustering method with a data-informed distance metric to aggregate [Cloudspotting](#) on Mars labels
- Derived a Martian cloud catalog from Mars Climate Sounder data, mapping seasonal and spatial distributions ([Icarus](#) 2024)

Amazon

Data Science Intern

Jan 2022 – May 2022
Luxembourg City, Luxembourg

- Built SQL ETL pipelines and live Python dashboards to monitor European transportation forecasts and detect anomalies
- Analyzed forecast trends across European countries, with a focus on how Brexit affected cross-border transportation

Airbus Defence and Space

Technical Program Management Intern

Oct 2021 – Jan 2022
Friedrichshafen, Germany

- Developed concepts for applying cloud computing and AI to data centers and autonomous systems
- Identified AI opportunities for decision-support use cases and assessed their technical and operational risks

European Space Agency (ESA)

Machine Learning Intern

Jul 2021 – Oct 2021
Noordwijk, Netherlands

- Designed and launched a citizen science project on [Hubble Asteroid Hunter](#), crowdsourcing labels on Hubble images
- Trained a deep learning classifier on them to measure satellite trails in 20 years of Hubble data ([Nature Astronomy](#) 2023)

German Aerospace Center (DLR)

R&D Intern

Jul 2020 – Oct 2020
Brunswick, Germany

- Designed a microgravity experiment for deployable space structures, flown on a NASA/DLR parabolic flight
- Built the experiment's LabVIEW control and force-measurement software and integrated its actuators and sensors

BMW Group

Data Science Intern

Jul 2019 – Sep 2019
Munich, Germany

- Built PySpark pipelines to prepare large datasets for downstream data science, with automated integrity checks
- Developed statistical methods to detect anomalies and outliers in the data, visualized in internal dashboards

SELECTED PUBLICATIONS

- [1] Padmavathi Venkatraman, Sydney Erickson, Phil Marshall, Martin Millon, Philip Holloway, Simon Birrer, **Steven Dillmann**, Xiangyu Huang, Sreevani Jaragula, Ralf Kaehler, Narayan Khadka, Grzegorz Madejski, Ayan Mitra, Kevin Reil, Aaron Roodman, and LSST Dark Energy Science Collaboration. Lens Model Accuracy in the Expected LSST Lensed AGN Sample. *The Astronomical Journal*, 172(4):223, 2026
- [2] Dingyuan Dai, Heli Qi, Lei Liu, ..., **Steven Dillmann**, ..., Juanzi Li, Weihao Xuan, and Tianyu Liu. OSWorld-Science: A Benchmark of Computer Use Agents for Learning and Using Scientific Software. *arXiv:2609.39903*, 2026
- [3] Lin Shi, Haowei Lin, Zixuan Zhu, ..., **Steven Dillmann**, ..., Mike Merrill, Ludwig Schmidt, and Alex Shaw. Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation. In *Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2026
- [4] Zijiao Chen, Nicholas Lu, Xinhui Li, Jocelyn A Ricard, Ce Ju, Huan H Wang, Christian Kindermann, Jeanette A Mumford, **Steven Dillmann**, James Kent, Alejandro de la Vega, Sanmi Koyejo, Vince D Calhoun, Joshua W Buckholtz, Juan Helen Zhou, Steffen Bollmann, and Russell A Poldrack. Bringing analytic rigor to agentic AI for science: The Brain Researcher platform for neuroimaging data analysis. *arXiv:2608.19902*, 2026
- [5] Negin Raoof, Richard Zhuang, Marianna Nezhurina, ..., **Steven Dillmann**, ..., Alex Dimakis, Benjamin Feuer, and Ludwig Schmidt. OpenThoughts-Agent: Data Recipes for Agentic Models. *arXiv:2606.24855*, 2026
- [6] Rishi Desai, Jesse Hu, Joan Cabezas, ..., Xiangyi Li, Ivan Bercovich, and **Steven Dillmann**. SWE-Marathon: Can Agents Autonomously Complete Ultra-Long-Horizon Software Work? *arXiv:2606.07682*, 2026
- [7] Garyk Brix, Matthew G Durrant, Jerome Ku, ..., **Steven Dillmann**, ..., Hani Goodarzi, Patrick D Hsu, and Brian L Hie. Genome modelling and design across all domains of life with Evo 2. *Nature*, 652(8112):1349–1361, 2026
- [8] Mike Merrill, Alexander Shaw, Nicholas Carlini, ..., **Steven Dillmann**, ..., Alex Dimakis, Andy Konwinski, and Ludwig Schmidt. Terminal-Bench: Benchmarking agents on hard, realistic tasks in command line interfaces. In *International Conference on Learning Representations (ICLR)*, 2026
- [9] Xiangyi Li, Yimin Liu, Wenbo Chen, ..., **Steven Dillmann**, Han-chung Lee, and Dawn Song. SkillsBench: Benchmarking how well agent skills work across diverse tasks. *arXiv:2602.12670*, 2026
- [10] Yuanqi Du, **Steven Dillmann**, Jon Laurent, Peter Jansen, Haojun Jia, Ludwig Schmidt, Andrew White, Kristin Persson, and Chenru Duan. Measuring AI Scientists: From Exams to Discovery. *ChemRxiv*, 2026
- [11] Christine Ye, Sihan Yuan, Suchetha Cooray, **Steven Dillmann**, Ian LV Roque, Dalya Baron, Philipp Frank, Sergio Martin-Alvarez, Nolan Koblishke, Frank J Qu, Diyi Yang, Risa Wechsler, and Ioana Ciuca. ReplicationBench: Can AI Agents Replicate Astrophysics Research Papers? *arXiv:2510.24591*, 2025
- [12] Trinidad Borrell, **Steven Dillmann**, Kshitij Duraphe, Furkan Eris, Kartheik Iyer, Ashod Khederlarian, Aman Kumar, Giovanni Marraffini, Michael J Smith, Shashwat Sourav, Rocco Di Tella, and John F Wu. The Platonic Universe: Do Foundation Models See the Same Sky? *arXiv:2509.19453*, 2025
- [13] Yanke Song, V Ashley Villar, Rafael Martínez-Galarza, and **Steven Dillmann**. A Poisson Process AutoDecoder for X-Ray Sources. *The Astrophysical Journal*, 988(1):143, 2025
- [14] Andrea Sacchi, Kevin Paggeot, **Steven Dillmann**, Juan Rafael Martínez-Galarza, and Peter Kosec. Hyperluminous Supersoft X-Ray Sources in the Chandra Catalog. *The Astrophysical Journal*, 983(2):124, 2025
- [15] **Steven Dillmann**, Juan Rafael Martínez-Galarza, Roberto Soria, Rosanne Di Stefano, and Vinay L Kashyap. Representation learning for time-domain high-energy astrophysics: Discovery of extragalactic fast X-ray transient XRT 200515. *Monthly Notices of the Royal Astronomical Society*, 537(2):931–955, 2025
- [16] Marek Slipiski, Armin Kleinböhl, **Steven Dillmann**, David M Kass, Jason Reimuller, Mark Wronkiewicz, and Gary Doran. The Cloudspotting on Mars citizen science project: Seasonal and spatial cloud distributions observed by the Mars Climate Sounder. *Icarus*, 419:115777, 2024
- [17] Sandor Kruk, Pablo García-Martín, Marcel Popescu, Ben Aussel, **Steven Dillmann**, Megan E Perks, Tamina Lund, Bruno Merín, Ross Thomson, Samet Karadag, and Mark J McCaughrean. The impact of satellite trails on Hubble Space Telescope observations. *Nature Astronomy*, 7(3):262–268, 2023

SELECTED HONORS & AWARDS

Stanford HAI Data Science Scholar	Fellowship for PhD research in data science and AI	Stanford HAI	2026
Stanford SIGF Fellowship	Fellowship for interdisciplinary doctoral research	Stanford University	2026
Head of Department Award	Top graduating student in Aeronautics	Imperial College London	2023
Best Computational Thesis Award	Best computational Master's thesis in Aeronautics	Imperial College London	2023
Turing Scheme Award	Funding for Master's thesis research at Harvard University	UK Government	2023
Engineering Leaders Scholarship	Prestigious UK engineering scholarship	Royal Academy of Engineering	2022
Ferry Porsche Prize	Top high-school graduate in physics, math, and technology	Porsche AG	2017

SELECTED TALKS

IFML Research Symposium on Machine Learning towards Superintelligence, Invited Talk	Oct 2026
<i>Evaluating AI Agents on Research Workflows Across Scientific Domains</i>	Austin, TX, USA
ICML 2026 Workshop on AI for Physics (AI4Physics), Invited Talk	Jul 2026
<i>Terminal-Bench-Science: Evaluating AI Agents on Scientific Research Workflows</i>	Seoul, South Korea
IAIFI Summer Workshop 2024, Contributed Talk	Aug 2024
<i>Data-Driven Discovery of X-ray Transients with Machine Learning</i>	Cambridge, MA, USA

ACADEMIC SERVICE

Organizer	Stanford Data Science Hackathon on Human-AI Research Replication
Reviewer	ICLR 2027, NeurIPS ML4PS Workshop 2025–2026, Agents4Science 2025, Astronomy & Computing